Fairness Preservation under Computational Budgets in Probabilistic Decision Systems

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 9; Team [*ID*]
[*Student 1 name and roll number*]; [*Student 2 name and roll number*]
Repository: `https://example.org/replace-with-repository`
Submission commit: `REPLACE`; Date: [*date*]

SMOKE RUN ON SYNTHETIC DATA — NOT EVIDENCE

Abstract

We ask whether a budget-aware fallback limits the fairness degradation that arises when exact inference in a nine-variable Bayesian network for pre-contact bank outreach is cut off by a deterministic operation budget. We compare fairness-aware fallback selection (validation worst-group decision deviation, AP loss ≤ 0.01) with a global fallback, utility-only selection and post-budget threshold adjustment. Primary result, paired block-bootstrap difference in Δ_{EOpp} against the global fallback: +0.000 [-0.125, +0.000] (Holm p=1.000). Limits: synthetic masks, no customer identifiers, small network.

1 Question, context and contribution

Outreach teams, customers, serving engineers and auditors are affected when a probabilistic scorer is run under a compute limit. $Y = 1$: recorded term-deposit subscription (UCI Bank Marketing); $D = 1$: selection for *hypothetical* outreach at fixed capacity. Only pre-contact information is used; call `duration` is excluded.

Research question: at a fixed operation budget, does selecting the fallback by validation worst-group deviation from the exact decision (subject to ≤ 0.01 AP loss) give a smaller Δ_{EOpp} than a global fallback, utility-only selection and threshold adjustment after budgeted inference? It is refuted if the Holm-adjusted paired interval for any difference includes 0 or is positive.

Contribution: a deterministic operation-count budget with evidence-dependent trips, a simulation isolating *group-dependent evidence availability* (with a no-dependence control), and the selection rule; elimination order never changes an exact posterior, only which queries exceed the budget.

Closest work: compute-budget disparities in neural inference [1]; our reproduced baseline is group-threshold post-processing [3]. Eight or more papers are compared in `docs/related_work.md`.

2 Data, graphical model and fairness method

Data and estimand: `bank-additional-full.csv` [2], date-ordered, no customer id; positional 60/20/20 split; all rows are contacts. Protected attributes: age band (< 30, 30–59, ≥ 60; primary) and marital status (audit only), crossed; groups with < 30 relevant observations are exploratory. Bins are fixed a priori; `unknown` values are treated as unobserved evidence.

Graph and inference: a Dirichlet-smoothed (train-only) discrete BN

$$p(\cdot) = p(A)\,p(E|A)\,p(J|A,E)\,p(F|J)\,p(H|A)\,p(L|H)\,p(P_d)\,p(P_r|P_d)\,p(O|P_r,P_d)\,p(Y|A,J,H,L,O,P_d),$$

with A age band, E education, J job, F default, H housing, L loan, P_d/P_r prior-contact bins, O previous outcome; no latent variables, the evidence mask R marks unobserved variables. Query $P(Y{=}1 \mid e)$ by exact variable elimination (min-fill, min-weight, min-degree) and clique-tree calibration, both with factor-entry operation counters; exact agreement: passed (max error 2.2e-16, tolerance 10^{-6}).

Method and fairness: a query whose exact cost would exceed the budget falls back to a precomputed table (prior, or conditioning on a variable subset). The proposed rule picks, on validation data, the candidate with the smallest worst-group decision-flip rate relative to exact inference among those

within the AP tolerance; routing is group-unaware. Primary criterion Δ_{EOpp} (range of group TPRs), chosen because the cost of a missed responder falls on the group that is never contacted; diagnostics: Δ_{DP}, Δ_{EOdds}, fallback-rate gap, flips, calibration. The age band is a model variable and thus used at prediction time.

3 Evaluation protocol

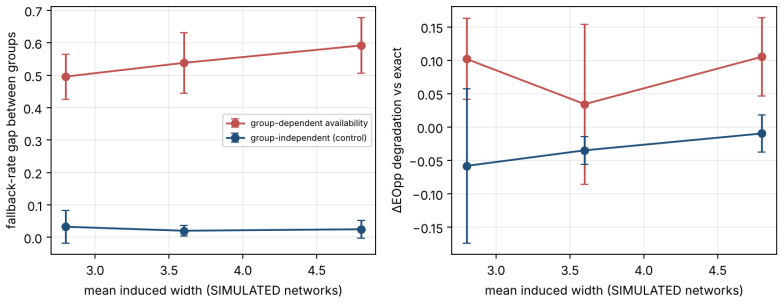
Baselines: exact inference (reference), global fallback, threshold adjustment after budgeted inference, fair-adjusted exact, logistic regression and gradient-boosted trees on the same variables. All use the same splits and information; thresholds, tables and selection use train/validation only and the test protocol is frozen (`docs/protocol.md`). Budgets are operation counts at the 50–100% quantiles of natural validation costs (primary: 75%, 11 operations); wall-clock is measured separately. Capacity 0.2. Workloads: natural (confirmatory) and synthetic group-dependent / group-independent masks (exploratory). Seeds: 5 for masks, networks and GBT. Intervals: fixed-model paired moving-block bootstrap over date-ordered test blocks (1000 replicates, 95%); Holm over the three confirmatory comparisons. Ablations: utility-only selection, global fallback, no evidence-based simplification, generous budget; robustness: elimination orders, densities, discretisation, capacities, masks.

4 Results and interpretation

SMOKE RUN ON SYNTHETIC DATA — NOT EVIDENCE

Method	AP ↑	Δ_{EOpp} ↓	Log loss / Brier ↓
Exact (reference)	0.229	0.309	0.288 / 0.078
Global fallback	0.212	0.258	0.292 / 0.079
Proposed (fair selection)	0.231	0.309	0.288 / 0.078
Utility-only selection	0.231	0.309	0.288 / 0.078
Threshold adj. after budget	0.212	0.171	0.292 / 0.079

Table note: natural workload, test size 1600, budget 11 operations, capacity 0.2; Δ_{EOpp} over groups with ≥ 30 positives. Paired proposed-minus-comparator effect on Δ_{EOpp} (negative favours proposed), 95% interval, Holm p: global fallback +0.000 [-0.125, +0.000] (Holm p=1.000); utility-only +0.000 [+0.000, +0.000] (Holm p=1.000); threshold adjustment +0.000 [+0.000, +0.048] (Holm p=1.000).



Main finding: claim an improvement only if the Holm-adjusted interval excludes 0 and AP loss stays ≤ 0.01; otherwise report a null result. [*After the real run: 2–3 sentences answering the research question with the estimates above.*]

Mechanism and failure: simulated networks (known structure, not facts about customers): group-dependent availability changes the fallback-rate gap by +0.516 on average versus 0.026 in the group-independent control; on the (synthetic-mask) real-data workload the gap contrast is +0.415 [+0.338, +0.493]. [*Name the ablation that most weakens the mechanism and one failure case from the main-results CSV.*]

5 Limitations, ethics and conclusion

Real evidence is the natural `unknown` pattern; masks and networks are simulations and their known effects are not population facts. Rows are not distinct customers (no identifiers), so intervals may be too narrow; groups below 30 observations are exploratory and wide intervals are not evidence of fairness. Historical contact selection means nothing is learned about never-contacted people, and an outreach rule that is fair across age and marital groups may still burden poorly documented customers. With nine variables exact inference is cheap, so budgets are defined relative to observed costs; conclusions need not transfer to denser networks. [*One-sentence bounded conclusion and next experiment (e.g. a denser network or real timing logs).*]

Reproducibility and individual contributions

Reproduce: `python scripts/run_experiments.py -mode real && python scripts/evaluate.py && python scripts/make_results.py`; configs: `configs/main.yaml`, `data.yaml`, `model.yaml`, `evaluation.yaml` Python 3.12, `requirements.txt`.
Student 1: [*artifacts and verification of Student 2's work*]. **Student 2:** [*artifacts and verification of Student 1's work*].
Evidence: `docs/requirements_matrix.md`, `results/`.

References

[1] S. Das, M. Romanelli, F. Fioretto. Disparate Impact on Group Accuracy of Linearization for Private Inference. ICML 2024. arXiv:2402.03629.

[2] S. Moro, P. Rita, P. Cortez. Bank Marketing. UCI Machine Learning Repository, dataset 222. `https://archive.ics.uci.edu/dataset/222/bank+marketing`; access date [*date*].

[3] M. Hardt, E. Price, N. Srebro. Equality of Opportunity in Supervised Learning. NeurIPS 2016. arXiv:1610.02413.